

Stochastic Processes

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Introduction

A family of random variables that is indexed by a parameter such as time is known as a **stochastic process** (or chance/random process).

Definition of Stochastic Process

Definition:

A stochastic process is a family of random variables

$$\{X(t) \mid t \in T\}$$

defined on a given probability space, indexed by the parameter t , where t varies over an index set T .

States and State Space

- The values assumed by the random variable $X(t)$ are called **states**.
- The set of all possible values forms the **state space** of the process.
- The state space is denoted by I .

Recall that a random variable is a function defined on the sample space S .

Thus, the stochastic process can be written as:

$$\{X(t, s) \mid s \in S, t \in T\}$$

This represents a family of functions depending on both s and t .

Random Variable at Fixed Time

For a fixed time $t = t_1$:

$$X_{t_1}(s) = X(t_1, s)$$

- This is a random variable as s varies over the sample space S .
- It is denoted by $X(t_1)$.

Similarly, for another time t_2 :

$$X_{t_2}(s) = X(t_2, s)$$

For a fixed sample point $s = s_1 \in S$:

$$X_{s_1}(t) = X(t, s_1)$$

- This is a function of time t .
- It is called a **sample function** or **realization** of the stochastic process.

Summary

- When t is fixed $\rightarrow$ we get a random variable.
- When s is fixed $\rightarrow$ we get a sample function.
- When both s and t vary $\rightarrow$ we obtain a stochastic process.

Queueing System Overview

Consider a computer server where:

- Jobs arrive randomly over time
- They wait in a queue if service is busy
- After service, they leave the system

Such systems can be modeled using stochastic processes.

Process 1: Number of Jobs After Departure

Let N_k = number of jobs in the system after the departure of the k -th customer.

Stochastic Process:

$$\{N_k \mid k = 1, 2, 3, \dots\}$$

- State space: $I = \{0, 1, 2, \dots\}$
- Index set: $T = \{1, 2, 3, \dots\}$

Type:

- Discrete-Time
- Discrete-State

Interpretation of N_k

- Observations are made only at departure instants
- Each step corresponds to one completed service
- Values change in integer steps

Example:

$$N_1 = 2, \quad N_2 = 3, \quad N_3 = 1, \dots$$

Process 2: Number of Jobs at Time t

Let $X(t)$ = number of jobs in the system at time t .

Stochastic Process:

$$\{X(t) \mid t \geq 0\}$$

- State space: $I = \{0, 1, 2, \dots\}$
- Index set: $T = [0, \infty)$

Type:

Consider the experiment of randomly choosing a resistor s from a set S of thermally agitated resistors and measuring the noise voltage $X(t, s)$ across the resistor at time t .

Sample functions for different resistors represent different realizations of the stochastic process.

Distribution Function at Fixed Time

At a fixed time $t = t_1$:

- Measure voltages across all resistors in S
- Count number of resistors with voltage $\leq x_1$
- Divide by total number of resistors

This gives the distribution function:

$$F_{X(t_1)}(x_1) = P(X(t_1) \leq x_1)$$

Distribution at Multiple Times

The same procedure can be repeated for:

$$t_2, t_3, \dots$$

to obtain:

$$F_{X(t_2)}(x), \quad F_{X(t_3)}(x), \dots$$

Thus, we obtain distributions of the process at different time instants.

Joint Distribution Function

The joint distribution of $X(t_1)$ and $X(t_2)$ is obtained by:

$$P(X(t_1) \leq x_1, X(t_2) \leq x_2)$$

Similarly, we can compute:

$$P(X(t_1), X(t_2), \dots, X(t_n))$$

- This defines the stochastic behavior of the process completely

Classification of Stochastic Processes

Based on State Space:

- **Discrete-State Process (Chain)** State space: $\{0, 1, 2, \dots\}$
- **Continuous-State Process** State space is continuous

Based on Time Parameter:

- **Discrete-Time Process** (Stochastic Sequence) $\{X_n \mid n \in T\}$
- **Continuous-Time Process**

Types of Stochastic Processes

	Discrete State	Continuous State
Discrete Time	Discrete-time Chain	Discrete-time Process
Continuous Time	Continuous-time Chain	Continuous-time Process

Problem Statement

Consider the stochastic process $\{S_k\}$, where S_k denotes the total number of successes in k independent tosses of a biased coin.

The probability of success (head) is:

$$p = 0.7$$

Generate 30 realizations of the process and estimate the mean and variance for:

$$k = 1, 5, 10, 50, 100$$

Mathematical Model

Each trial is represented as a random variable:

$$X_i = \begin{cases} 1, & \text{if success (head)} \\ 0, & \text{if failure (tail)} \end{cases}$$

Thus, the process is:

$$S_k = X_1 + X_2 + \cdots + X_k$$

Since trials are independent:

$$S_k \sim \text{Binomial}(k, 0.7)$$

Interpretation of the Process

- S_k represents cumulative successes after k trials
- The process evolves as k increases
- Each realization gives one possible outcome path
- 30 realizations $\rightarrow$ 30 different sample paths

Theoretical Mean

For binomial distribution:

$$E[S_k] = kp$$

Substituting $p = 0.7$:

$$E[S_k] = 0.7k$$

Interpretation:

- On average, 70% of trials result in success
- Mean increases linearly with k

Theoretical Variance

For binomial distribution:

$$\text{Var}(S_k) = kp(1 - p)$$

Substituting $p = 0.7$:

$$\text{Var}(S_k) = 0.7 \times 0.3 \times k = 0.21k$$

Interpretation:

- Variability increases with k
- Spread grows but in a predictable way

Simulation Procedure

- Generate 30 independent realizations of the process
- For each value of k , compute:
 - Sample mean:

$$\hat{E}[S_k] = \frac{1}{30} \sum_{i=1}^{30} S_k^{(i)}$$

- Sample variance:

$$\widehat{\text{Var}}(S_k)$$

Observed Results (Typical)

k	Mean (Est.)	Mean (Theory)	Var (Est.)	Var (Theory)
1	0.7	0.7	0.21	0.21
5	3.4–3.6	3.5	1.0–1.2	1.05
10	6.8–7.2	7	2.0–2.4	2.1
50	34–36	35	9–12	10.5
100	68–72	70	18–23	21

Graphical Interpretation

- Mean grows linearly with k
- Variance also grows linearly
- Sample paths become smoother for large k

Observation:

- Relative fluctuation decreases as k increases

Key Observations

- Estimated mean $\approx$ theoretical mean $0.7k$
- Estimated variance $\approx$ theoretical variance $0.21k$
- Accuracy improves with larger k

Conclusion

- The process follows a binomial distribution
- Simulation validates theoretical results
- Demonstrates:
 - Law of Large Numbers
 - Central Limit Theorem
- For large k , distribution becomes approximately normal

Problem Statement

Consider a stochastic process $\{S_k\}$, where S_k denotes the total number of customers arriving in k independent time slots.

In each time slot, a customer arrives with probability:

$$p = 0.4$$

independently of others.

For each fixed k , use the observed values of S_k to estimate the mean and variance.

Generate 30 realizations and estimate the mean and variance for:

$$k = 1, 5, 10, 50, 100$$

Step 1: Define the Process

$$X_i = \begin{cases} 1, & \text{arrival} \\ 0, & \text{no arrival} \end{cases}$$

$$S_k = X_1 + X_2 + \cdots + X_k$$

$$S_k \sim \text{Binomial}(k, 0.4)$$

Step 2: Theoretical Values

$$E[S_k] = 0.4k$$

$$\text{Var}(S_k) = 0.4 \times 0.6 \times k = 0.24k$$

Step 3: Estimated Mean

Let the 30 observed values be:

$$S_k^{(1)}, S_k^{(2)}, \dots, S_k^{(30)}$$

Estimated mean is:

$$\hat{E}[S_k] = \frac{1}{30} \sum_{i=1}^{30} S_k^{(i)}$$

Interpretation:

- Average number of arrivals across all realizations

Step 4: Estimated Variance

Step 1: Compute deviations

$$S_k^{(i)} - \hat{E}[S_k]$$

Step 2: Square deviations

$$(S_k^{(i)} - \hat{E}[S_k])^2$$

Step 3: Take average

$$\widehat{\text{Var}}(S_k) = \frac{1}{30} \sum_{i=1}^{30} (S_k^{(i)} - \hat{E}[S_k])^2$$

Step 5: Example Calculation (for $k = 5$)

Suppose observed values are:

$$2, 1, 3, 2, 2$$

Mean:

$$\hat{E}[S_5] = 2$$

Variance:

$$\widehat{\text{Var}}(S_5) = \frac{(0^2 + (-1)^2 + 1^2 + 0^2 + 0^2)}{5} = 0.4$$

Step 6: Observed Results

k	Mean (Est.)	Mean (Theory)	Var (Est.)	Var (Theory)
1	0.4	0.4	0.24	0.24
5	2	2	1.2	1.2
10	4	4	2.4	2.4
50	20	20	12	12
100	40	40	24	24

Conclusion

- Estimated mean $\approx 0.4k$
- Estimated variance $\approx 0.24k$
- Results improve as number of realizations increases
- Demonstrates:
 - Law of Large Numbers
 - Central Limit Theorem